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# Kinetic Mining in Context : Few-Shot Action Synthesis via Text-to-Motion Distillation

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## 1 Motivations

**Context** — Skeletal Human Activity Recognition (HAR) is key for applications like surveillance, robotics, and sports, but performance is heavily constrained by the **high cost of collecting annotated large amount of motion data** [11].

**Approach in brief** — **Generative data augmentation** offers a compelling solution by synthesizing new training samples [11]. Recently, **Text-to-Motion (T2M) diffusion models** [31] have emerged as powerful generative priors, trained on large text-annotated datasets.

**The problem / research gap** — However, using T2M models for HAR reveals a **domain gap**:

- **T2M** priors generate broad, general, fluid movements from free-form text...
- But **HAR** requires short, kinematically precise actions tied to discrete labels, causing direct transfer to fail.

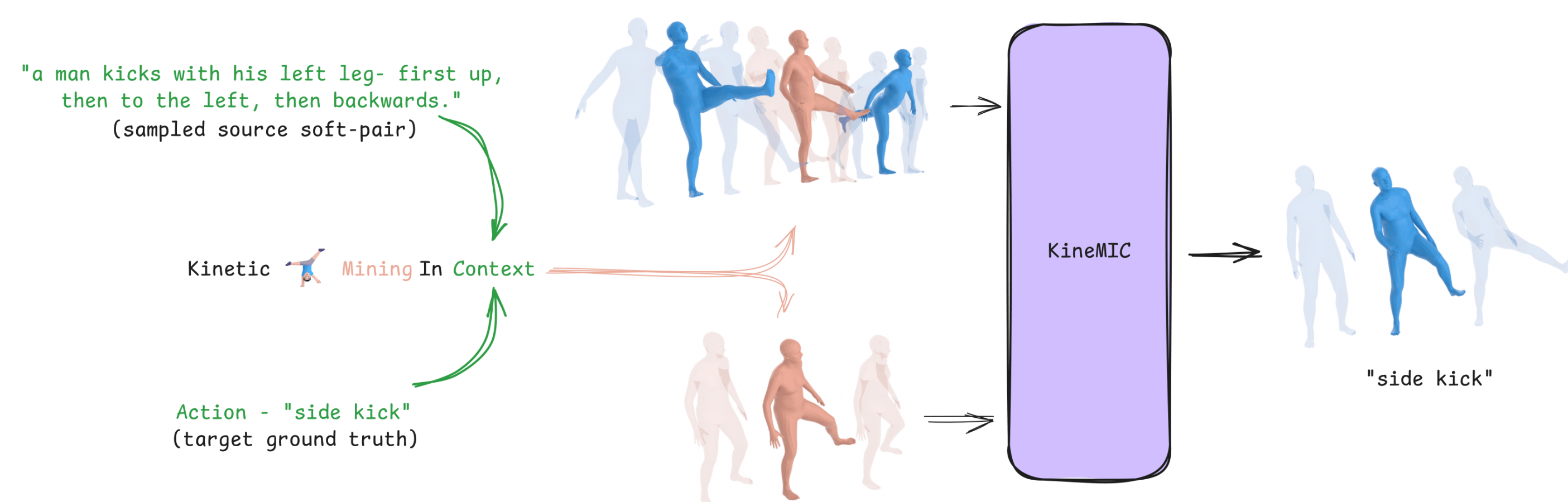
**Objective** — We propose Kinetic Mining in Context (KineMIC), a teacher-student framework that adapts generalist T2M diffusion models into specialized Action-to-Motion generators, enabling high-quality few-shot synthetic data augmentation.

## 2 Method

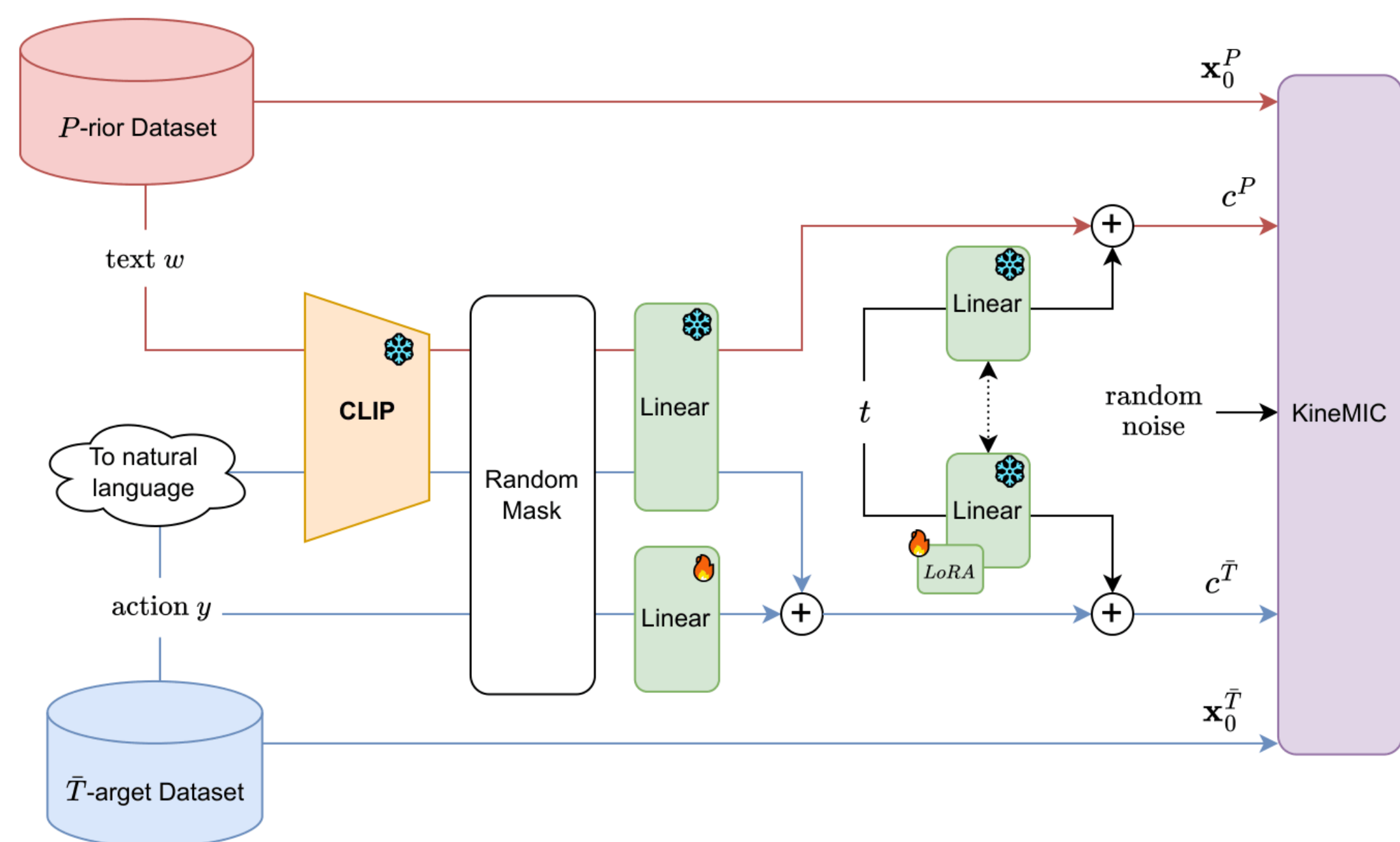
We work with two datasets: a **large T2M dataset**, and a **tiny HAR dataset** with just **10 samples per action class**. T2M datasets are not built for HAR, but because of their scale motion patterns we need can likely be found (**mined**) within them.

We use **CLIP** [25] to match action labels with text prompts, assuming text similarity acts as a reasonable proxy to kinematic similarity and perform **paired training** on the 2 datasets.

💡 **Our few HAR samples drive the mining process** 💡 KineMIC uses them as context to dig into **long source sequences** and extract only the **relevant kinematic snippet**.

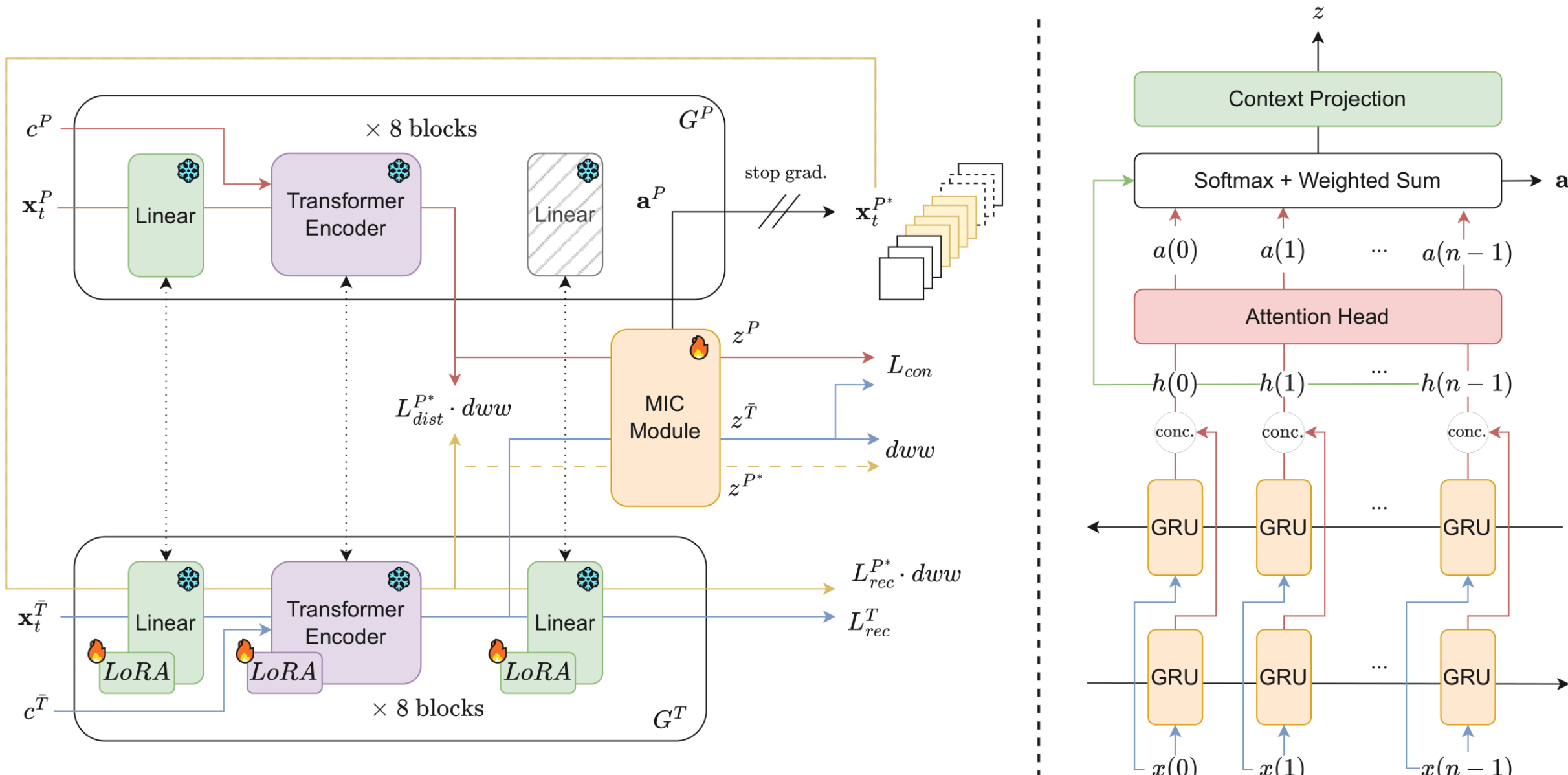


We consider MDM [31] as our core model. The **prior stream** (red) processes T2M data from **HumanML3D** [13] producing conditioning  $c^P$  driven by text only. The **target stream** (blue) processes **A2M** data from **NTU-RGB+D 120** [19], producing conditioning  $c^T$  driven by both text and action id.



**Left** — A frozen teacher  $G^P$  (top) and LoRA-tuned student  $G^T$  (bottom) process noisy motions.

**Right** — The Mining In Context (MIC) module uses an **attention-enhanced bidirectional-GRU** to aggregate frame tokens into latent  $z$  vectors. The frame sequence maximizing the cumulative attention, which we call the **"prior window"**  $x_t^{P*}$ , becomes **additional training** for the student!



## 3 Model Training

**Contrastive Alignment** ( $L_{con}$ ): Trains the MIC module using Soft Nearest Neighbors loss [10] to push motion pairs latent in a shared space.

**Reconstruction Losses** ( $L_{rec}^T, L_{rec}^P$ ): A mean squared error forces student model  $G^T$  to accurately reconstruct both real target motions and mined prior windows under target action conditioning.

**Window Distillation** ( $L_{dist}^P$ ): A mean square error enforces internal motion representations from frozen teacher  $G^P$  to student  $G^T$  across the mined frame window.

**Dynamic Window Weighting** ( $dww$ ): Scales prior reconstruction and distillation losses by cosine similarity score to down-weight low-quality mined windows... That's very important! We don't have guarantees on the quality of pairs (they're computed once before training starts through CLIP).

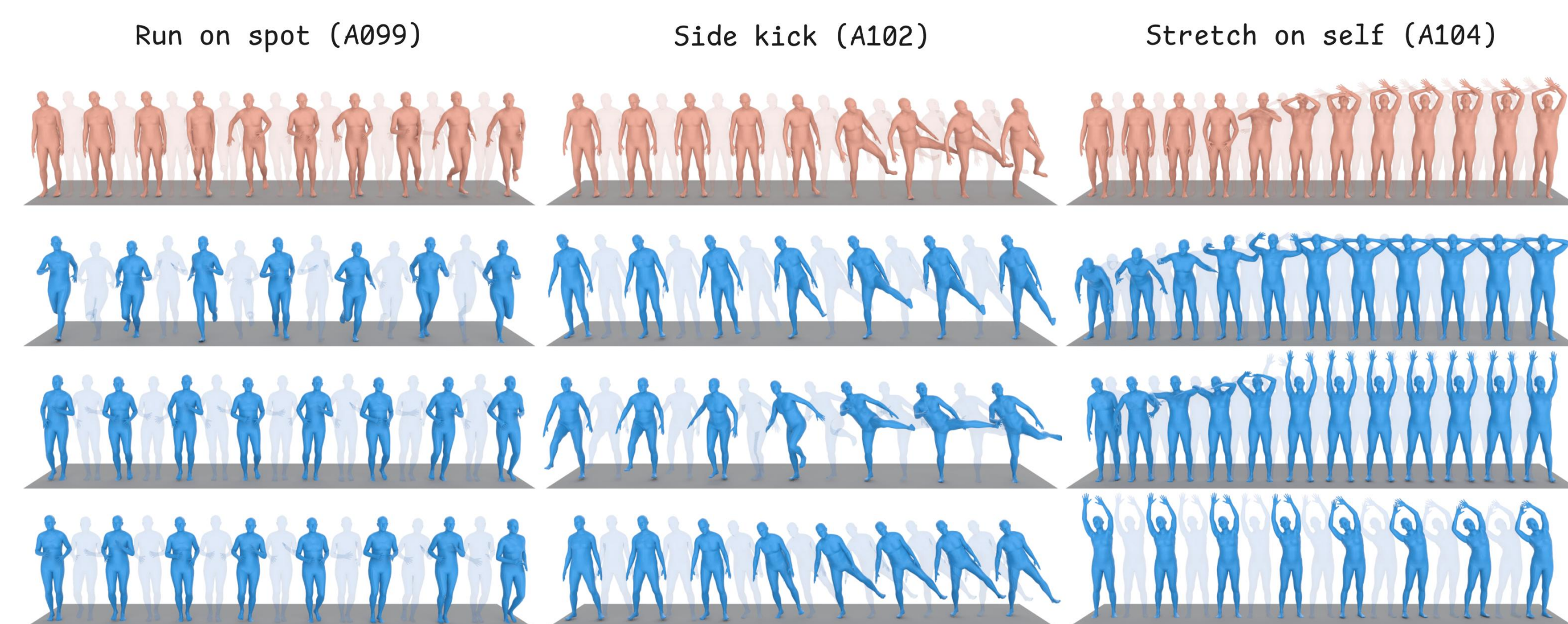
$$\mathcal{L} = \mathcal{L}_{rec}^T + \lambda_{con} \mathcal{L}_{con} + dww \cdot (\mathcal{L}_{rec}^P + \lambda_{dist} \mathcal{L}_{dist}^P)$$

## 4 Results

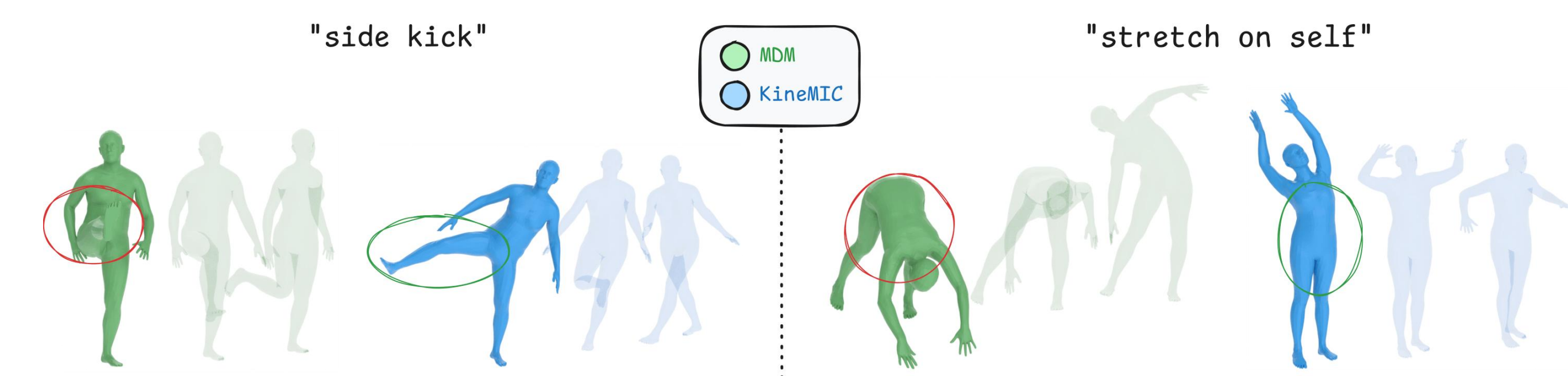
We evaluate on three action classes ("**side kick**", "**stretch on self**", and "**run on spot**") to **test how synthetic data boosts downstream ST-GCN [33] recognition accuracy**. Our framework matches prior state-of-the-art performance using a fundamentally different strategy, **showing that diffusion Text-to-Motion priors can be effectively repurposed for few-shot HAR synthesis**.

| Source                  | Method                                   | Top-1 Acc (%) ↑ |
|-------------------------|------------------------------------------|-----------------|
| Prior Work <sup>†</sup> | Real-data-only (30 samples) <sup>†</sup> | 58.4            |
|                         | ACTOR <sup>†</sup> [22]                  | 73.6            |
|                         | Kinetic-GAN <sup>†</sup> [4]             | 81.7            |
|                         | Fukushi et al. <sup>†</sup> [11]         | 86.4            |
| Our Analysis            | Real-data-only (30 samples)              | 63.1            |
|                         | MDM (pre-trained)                        | 83.9            |
|                         | MDM (from scratch)                       | 73.9            |
|                         | MDM (LoRA fine-tune)                     | 75.5            |
|                         | KineMIC                                  | 86.2            |

**KineMIC** synthesizes **diverse motions** from just a **few-shot samples**, remaining both semantically accurate and faithful to the ground-truth kinematic signature.



Text-only conditioning with MDM [31] fails to synthesize accurate class kinematics, misinterpreting a "side kick" prompt as a standard front kick. By grounding generation in few-shot NTU-RGB+D 120 [19] context, our model ensures high variety and kinematic fidelity despite very limited data.



## Limitations & Future Work

**Semantic Proxy Bottleneck:** Building the motion pairs through CLIP relies on text similarity, which can retrieve kinematically inaccurate samples (e.g., retrieving a "front kick" for a "side kick" action).

**Filtering Dependency:** Because text retrieval is imperfect, performance relies heavily on Dynamic Window Weighting ( $dww$ ) to filter out noisy source windows.

**Future Directions:** Testing on broader, more complicate motion datasets and integrating prompt augmentation strategies to improve semantic retrieval quality.

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